Wireless Sensor Networks: P3 Exercises **26/10/2026**

For the following exercises use the C source files of client and server (for example: p3_udp_client.c, p3_udp_server.c), based on the procedure followed in the lab examples.

Based on that, proceed with the following exercises:

1. Start from the client-server communication example, and whenever the server (root) receives a packet from a client (sensor node) generate a respond packet back to the client. The client must show in its terminal (via serial line) the content of the received packet.

2. Send periodically sensor data (temperature and humidity values) from the clients to the server and show that data in the terminal line of the server.

3. Based on the exercises carried out in P2, send the actions commands from the server (root) to the clients (sensor nodes) to request/configure measurements from the sensors and activate remotely the LEDs. Use the same frame format defined in P2 in combination with the examples of the client-server communication.

Evaluation

The exercises will be evaluated through a document that should include the following items:

- Code of each exercise (inside the document).
- Brief explanation with screen captures showing the results of the exercise.
- Upload this document to **Moodle** by Friday the **10th of November** (a specific homework task will be created in Moodle for it).
- Remember this is part of the evaluation of the subject.